Secure Text and Voice Chat Implementation Using RING- LWE and ChaCha20 Cryptography

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***Abstract*— There is a always a need to safeguard the data and maintain secure sensitive data transmission by avoiding quantum attacks and emerging threat, various RSA and ECC become vulnerable to quantum attacks. To avoid this Lattice based cryptography like Ring Learning with Errors (RING - LWE)and ChaCha20 can be used to ensure strong security against classical and quantum attacks, so that we can ensure a secure data transmission.**

Index Terms—Ring Learning with errors (RING – LWE, RLWE), ChaCha20, Learning with errors (LWE)

I. Introduction

There is always a need to safeguard data and maintain secure sensitive data transmission by avoiding quantum attacks and emerging threats. Various RSA and ECC methods become vulnerable to quantum attacks. To address this, lattice-based cryptography like Ring Learning with Errors (RING-LWE) can be used to ensure strong security against classical and quantum attacks, enabling secure data transmission.

Emerging technologies highlight the need for lattice cryptography. The problem is that many existing methods like RSA, Diffie-Hellman, or ElGamal will be broken, and their security will be severely reduced with the advent of quantum computers. Thus, we need to quickly move from existing methods to quantum-robust cryptographic techniques.

Traditional code-based cryptography relies on factors like error-correcting codes. Typically, symmetric key encryption, such as AES, is used for core data encryption, while public key encryption is used for signing a hash and for key exchange. For example, when we have some data, we generate its hash, encrypt the hash using the sender's private key, and decrypt it using the public key. Since both parties share the same key, they create a shared secret. This method, as used in algorithms like Diffie-Hellman, is now easier for attackers to break.

II. LITERATURE SURVEY

Tuy Nguyen Tan et al. (2019) introduced a high-secure fingerprint authentication system utilizing Ring-LWE cryptography to enhance the protection of fingerprint data. The authors proposed a delay-optimized fingerprint feature extraction approach combined with a Ring-LWE cryptographic scheme employing low-latency NTT-based polynomial multiplications. The dataset included public databases from FVC competitions (FVC2000, FVC2002, FVC2004), and results showed a 46% improvement in encryption time and a 44% improvement in decryption time compared to existing schemes.

Yushu Yang et al. (2024) introduced a lightweight and efficient encryption/decryption coprocessor for RLWE-based cryptography. The study proposed enhancements to the Schoolbook Polynomial Multiplication (SPM) algorithm, reducing its time complexity from n2n^2n2 to n2/8n^2/8n2/8 and incorporating a Compressed CDT Gaussian sampler for efficient key generation. The proposed coprocessor demonstrated a 2.3× performance improvement and 2.7× hardware efficiency for encryption, and a 2.4× performance speedup with 3.2× hardware efficiency for decryption, compared to state-of-the-art solutions.

III. Lattice-based Cryptography

Lattice-based cryptography is the generic term for constructions of cryptographic primitives that involve lattices, either in the construction itself or in the security proof. Lattice-based constructions support important standards of post-quantum cryptography [3].

A. In what way Lattice based cryptography is better

Unlike more widely used and known public-key schemes such as the RSA, Diffie-Hellman or elliptic-curve cryptosystems — which could, theoretically, be defeated using Shor's algorithm on a quantum computer — some lattice-based constructions appear to be resistant to attack by both classical and quantum computers. Furthermore, many lattice-based constructions are considered to be secure under the assumption that certain well-studied computational lattice problems cannot be solved efficiently[3].

Using lattice methods, we can ensure greater security. By employing vectors with more than 100 dimensions, the many possible combinations form a lattice. The points on this lattice are called lattice grids. We generate a point on the lattice, introduce some form of error to create a new point, and generate a new vector. The challenge lies in finding the closest vector to the actual point, which is the essence of lattice cryptography.

In two-dimensional lattices, finding the original point is easier, but in multidimensional lattices, it becomes difficult without additional information about lattice points. There are also other vectors closer to the origin point, leading to a problem called the Closest Vector Problem (CVP).

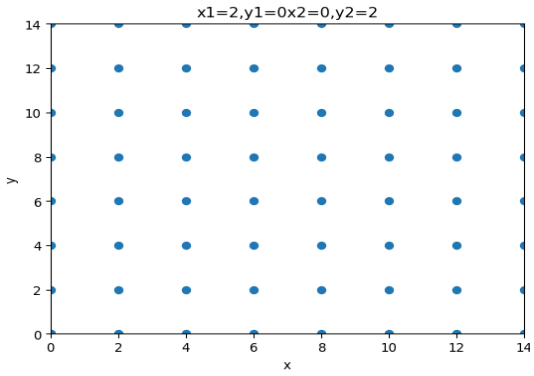
B. *Clocest Vector Problem (CVP)*

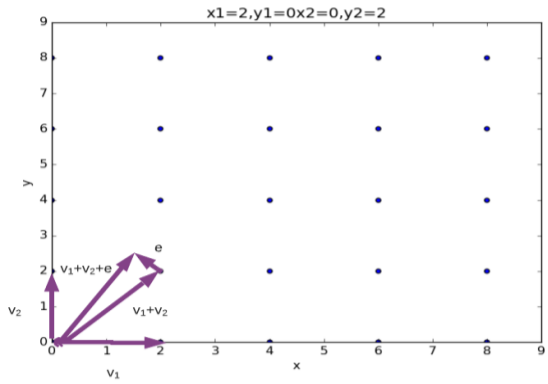
is the computational problem: given a basis matrix B for L and a vector w ∈ Q^m (one can work with high-precision approximations in R^m, but this is essentially still working in Q^m), compute v ∈ L such that || w – v || is minimal.

C. *Lattice Plot*

In lattice-based cryptography we create a lattice structure within a number of dimensions, and then pick a point, and define this as our vector.[4] We then add a small error onto this vector, and the difficult problem is then to find the original vector within the lattice:

(for parameters x1= 2, y1= 0, x2= 0, y2= 2)





while finding a short vector and the closest point looks easy on our grid

The difficulty in lattice cryptography is introduced with a small noise vector ($\vec{e}$ e→) onto our original vector, and where it is then difficult to find what the actual vector is for our defined lattice [4]

D. *Learning With Errors (LWE)*

The LWE works by taking a matrix A, a secret matrix S, and an error matrix E. The problem is represented as:

$$Z^{(mxn)}(k)$$

When we calculate A x S + E, we perform a row-column operation and apply a modular operation with a number

Real-time examples often use a polynomial approach with three polynomials: a random polynomial, a secret polynomial, and a small noise polynomial:

B= A x S(A) + E(A).

E. *Advantages of using LWE methods*

- It is hardest problem, as it delivers worst case lattice problems, which are even more hard
- The LWE approach is versatile
- We obtain a algorithm with running time

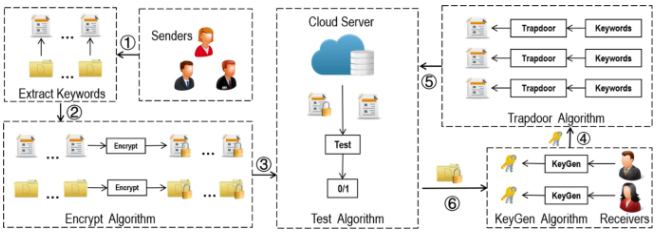
$$q^{n} = 2^{O(\text{nlogn})}$$

  Using only O(n) equations.
- Algebraic lattice problems are easy, whereas the geometric lattice problems are hard

IV. RING – LWE

Ring-lwe is used to create lwe based homomorphic encryption schemes more efficient, using this scheme we can pack more information to a plain text so encryption becomes more efficient. It is a Resource efficient cryptographic scheme, which is appropriate for lightweight systems

A. *RLWE architecture*



B. *RLWE working*

Ring LWE is a key exchange algorithm. let's say a client and server exist

client generates its secret as (Sb)

server generates its secret as (Sa).

A is a shared polynomial among the network., For every connection, they create a new shared secret to generate the same shared key.

The server computes

$bA = sa \times A + e\,(a)$

The client computes

$aB = sb \times A + e\,(b)$

They exchange b A and a B,

Both sides multiply the received value by their respective secrets (s a , s b) to compute the shared key

it passess over they exchange the values of bA and aB, and they multiply that with their secret values sa, sb

Sh (server) = ((sb) x A + e(b)) x (sa)

Sh (client) = ((sa) x A + e(a)) x (sb)

The shared key generation uses probabilistic encryption to remove errors:

Sh(server) = ((sb) × (sa) × A + e(b) × sa),

Sh(client) = ((sa) × (sb) × A + e(a) × sb).

Here e(a)×sb and e(b)×sa are noise/error values. Probabilistic encryption removes these errors. In most cases, the noise is much smaller than the overall value, making it easy to filter out the error values.

Classical set up: n is a power of 2; q is a prime, q = 1 (mod 2n).

R := Z[x]/(x + 1)

ao(x)s(x) + e0(x) = b0(x) mod qR

a1(x)s(x) + e1(x) = b1(x) mod qR

where a;(x) are uniformly random polynomials, and e;(x) are unknown small random polynomials.

given a secret message as a polynomial in this ring and we have to solve the secret polynomial using a random noisy system

ao(x)s(x) + e0(x) = b0(x) mod qR
a1(x)s(x) + e1(x) = b1(x) mod qR
. . . .
. . . .
.. ..

the situation is same as LWE, but here we have the random co-efficient a(x) which is sampled uniformly from this ring R and we have small secret polynomial Ei(x).

C. *ChaCha20 Algorithm: Encryption and Decryption*

ChaCha20 is a performant and lightweight, yet highly secure, 256-bit stream cipher used to encrypt and decrypt data.

Algorithm : The encryption and decryption of voice using ChaCha20

Input:

Key: K (256 bits)
Nonce: N (64 bits)
Plaintext: m ∈ $\{0,1\}^n$

Output:

Encrypted ciphertext: C
Decrypted plaintext: m′

1: Initialize state with, **Key K, nonce N and counter ctr**
2: Generate keystream **stream=ChaCha20(K,N,ctr).**
3: Encryption **c = m ⊕ stream**.
4: Decrypting **m′ = c ⊕ stream**.
5: Return **C** for encryption or $\boldsymbol{m}'$ for decryption.

Encryption: The plaintext is XORed with the keystream generated from ChaCha20's state to create ciphertext.

Decryption : he same XOR operation is used to recover the original plaintext from the ciphertext, as the same keystream is generated during encryption and decryption

D. *RLWE Algorithm : Key Generation with Encryption and Decryption*

The process of key generation, encryption and decryption in RLWE is [1]

Algorithm : The Key Generation using RLWE

Input:

Shared Polynomial: A
Client Secret: Sb
Server Secret: Sa
Output: Shared Key: Sh

1. Server computes Ba= Sa • A + ea
2. Client computes Ab = sb• A + eъ
3. Server sends Ba to the client, and the client sends Ab to the server
4. Server computes Sh server = Sa•Ab
5. Client computes Sh client = sb.Ba
6. Sherver = Shelient (with probabilistic noise correction)
7. Return Sh

Algorithm : The encryption and decryption of RLWE

Input:

Public key: $\boldsymbol{p} \leftarrow R$ and $\boldsymbol{a} \leftarrow R$;
Secret key: $\boldsymbol{r2} \leftarrow D\sigma$; Plaintext: $\boldsymbol{m} \leftarrow \{0,1\}^{n}$.

Output: Decrypted plaintext: $\boldsymbol{m}'$.

1: Choose the noise terms $\boldsymbol{e1}, \boldsymbol{e2}, \boldsymbol{e3} \leftarrow \boldsymbol{D\sigma}$;
2: Encode plaintext $\overline{\boldsymbol{m}} = \boldsymbol{ENCODE}(\boldsymbol{m})$;
3: Encryption $\boldsymbol{c1} = \boldsymbol{ae1} + \boldsymbol{e2}$, $\boldsymbol{c2} = \boldsymbol{pe1} + \boldsymbol{e3} + \overline{\boldsymbol{m}}$;
4: Decrypting $\boldsymbol{c} = \boldsymbol{c1} \times \boldsymbol{r2} + \boldsymbol{c2}$;
5: Decode plaintext $\boldsymbol{m}' = \boldsymbol{DECODE}(\boldsymbol{c})$;
6: return $\boldsymbol{m}'$

.

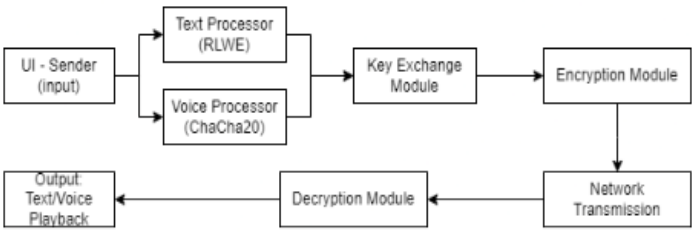
the multibit plaintext m is transformed into a polynomial $\overline{m}$ by the encoding module

$$\overline{m} = ENCODE(m) = \frac{(q-1)}{2} \text{ x m}$$

The plaintext c after decryption is transformed into bit plaintext m' by the decoding unit

E. *Architecture diagram of RLWE Proposed implementation*

The Proposed implementation is Secure voice and text application using Ring LWE cryptosystem.



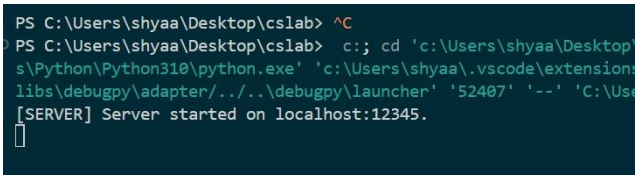
the text and voice are the inputs captured, the text data is pre-processed, and the voice data is digitized and compressed. the Ring LWE algorithm securely generates and shares secret keys between sender and receiver. the shared key is then used to encrypt the text or voice data. encrypted data is transmitted over the network and sent to decryption. the received data is then decrypted by using the shared key and displays the text or voice message in the application

F. Proposed Implementation of RLWE in Secure voice and Text Application

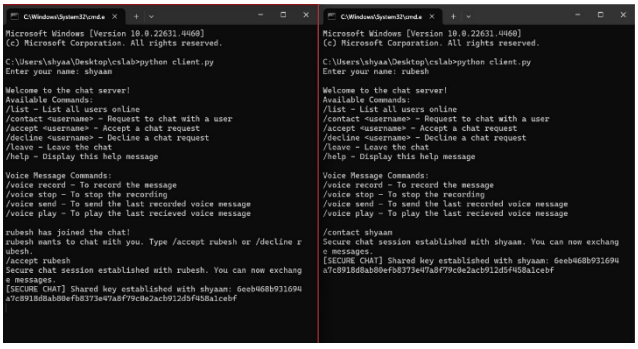
The Chat application with text and voice message facility is developed using secured data transmission and highly secured encryption standards. The Ring Learning with error algorithm is used to generate and share the key with that key encryption and decryption is handled by applying chacha20 encryption and decryption algorithm

RESULT

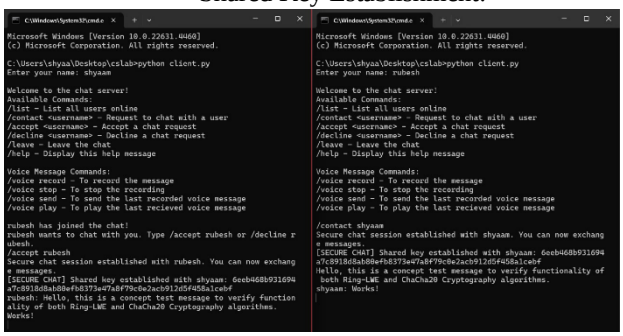
Thus the Proposed Encryption and Decryption using Ring-LWE and ChaCha20 was implemented the proposed encryption scheme maintains quality of text and audio throughout the process. process. By the use of keys, we can ensure that no other third party can decrypt the message and get the original message.



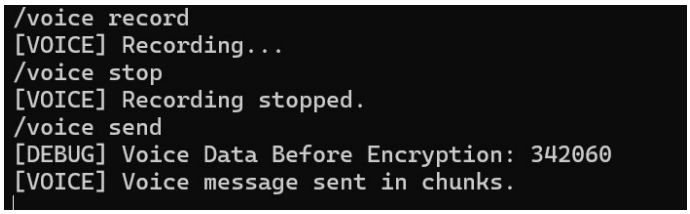
Server on Initiation



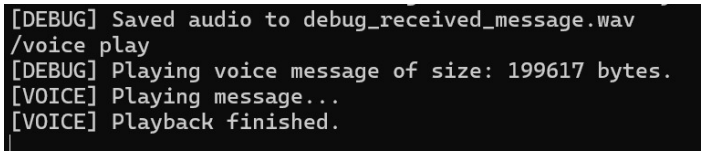
Shared Key Establishment:



message test



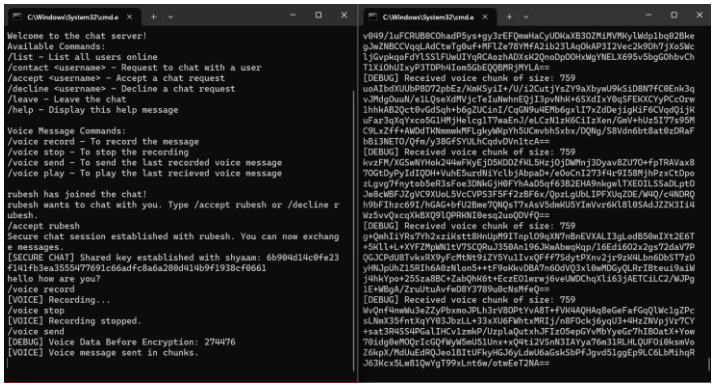
Sender sending voice message



Recipient side voice message

Encrypted messages at server

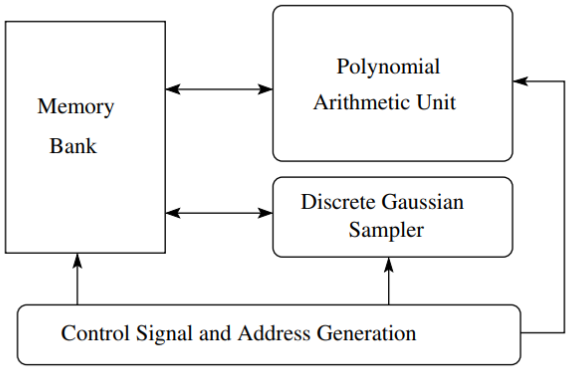
Server Log



Voice Transmission

NOVELTY

By combining Ring LWE and .ChaCha20 we have came up with more secure method of Messaging application



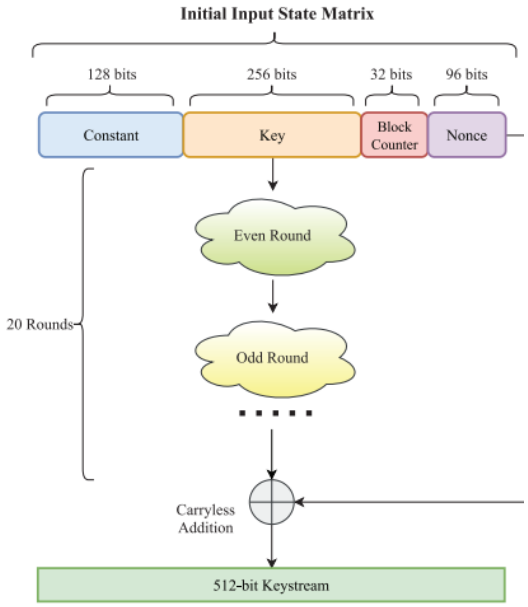


FIGURE 5. ChaCha20 algorithm flow.

FUTURE WORKS

Prove the hardness of RLWE problem underlying real-world HE schemes, like the ones used in Microsoft's SEAL or IBM's HElib.[5]
Assumptions: cyclotomic fields; non-dual ring; discrete and small error; small secret; split modulus.
Idea: RLWE mod q decision-RLWE. Show RLWE mod q is secure by showing e mod q is pseudorandom.
Two possible ways: (1) Study Regev's smoothing parameters. (2) Use Fourier analysis and properties of the Jacobi function theta

CHALLENGES

Poor efficiency in encrypting and decrypting big data and data formats other than text in Ring LWE

Realtime Encryption cannot be carried over with Ring LWE

In lattice cryptography if there more than one vector adjacent to each other vectors closer to the origin point, leading to a problem called the Closest Vector Problem (CVP).

This method is inefficient on encrypting a single audio file as it can be efficient when data is sent as chunks

Voice messages are recorded and sent as files, not streamed in real-time in this method of implementation.

CONCLUSION

A Secure chat application was implemented as proposed by Ring LWE and ChaCha20. The integration of these modules were implemented by incorporating the Ring LWE and ChaCha20 which in turn helped in encrypting and decrypting a text or voice message sent through our secured messaging application.

the project ensures secure communication and also increases error handling with authentication. further real-time voice streaming also have been tried in this project. produces a novelty of integrating two security standards

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